

Theoretical and Computer Exercise: Extended Kalman filter for a pendulum

Problem 1: Theoretical Preparation

Consider a continuous-time nonlinear simple pendulum system subject to random torque disturbances. The state vector $\mathbf{x}(t) = \begin{bmatrix} \theta(t) \\ \omega(t) \end{bmatrix}$ represents the angle θ (in rad) and the angular velocity ω (in rad/s). The system dynamics are:

$$\begin{aligned}\dot{\theta}(t) &= \omega(t) \\ \dot{\omega}(t) &= -\frac{g}{L} \sin(\theta(t)) - \frac{d}{m} \omega(t) + w(t)\end{aligned}$$

where $g = 9.81 \text{ m/s}^2$ is gravity, $L = 1 \text{ m}$ is the pendulum length, $m = 1 \text{ kg}$ is the mass, $d = 0.5 \text{ Ns/m}$ is the damping coefficient, and $w(t)$ is the process noise. The horizontal position of the pendulum bob is measured, which is a nonlinear function of the angle:

$$y(t) = L \sin(\theta(t)) + v(t)$$

where $v(t)$ is the measurement noise.

- Write the continuous-time system in the standard nonlinear state-space form $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) + \mathbf{\Gamma}w(t)$ and $y(t) = h(\mathbf{x}(t)) + v(t)$.
- Linearize the continuous-time dynamics $\mathbf{f}(\mathbf{x})$ and measurement $h(\mathbf{x})$ around the stable downward equilibrium point $\mathbf{x}_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ to find the continuous-time matrices $\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}_e}$ and $\mathbf{C} = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\mathbf{x}_e}$. This represents a standard Linear Kalman Filter approach that assumes small angles.
- Discretize the nonlinear system using the forward Euler method with a sampling time of $T_s = 0.005 \text{ s}$. Define the discrete-time nonlinear state transition function $\mathbf{x}_k = \boldsymbol{\varphi}(\mathbf{x}_{k-1}) + \mathbf{\Gamma}_d w_{k-1}$ and discrete measurement $y_k = h(\mathbf{x}_k) + v_k$.
- The Extended Kalman Filter (EKF) requires the Jacobians of the discrete-time system evaluated at the current state estimate $\hat{\mathbf{x}}$. Compute the Jacobians $\Phi = \frac{\partial \boldsymbol{\varphi}}{\partial \mathbf{x}}$ and $\mathbf{H} = \frac{\partial h}{\partial \mathbf{x}}$ as functions of an arbitrary state $\mathbf{x} = \begin{bmatrix} \theta \\ \omega \end{bmatrix}$.
- Perform a manual computation of one EKF prediction and update step. Assume the current estimate is $\hat{\mathbf{x}}_0^{(+)} = \begin{bmatrix} \pi/2 \\ 0 \end{bmatrix}$ with covariance $\mathbf{P}_0^{(+)} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}$. Use the numerical parameters $g = 9.81, L = 1, m = 1, d = 0.5, T_s = 0.005, Q = 0.1, R = 0.05$. Assume a new measurement $y_1 = 1.05$ is received. Calculate $\hat{\mathbf{x}}_1^{(-)}, \mathbf{P}_1^{(-)}$, and the Kalman gain $\mathbf{K}_1$. What do you notice about the Kalman gain in this specific state, and why?